Discussion Section Worksheet 4

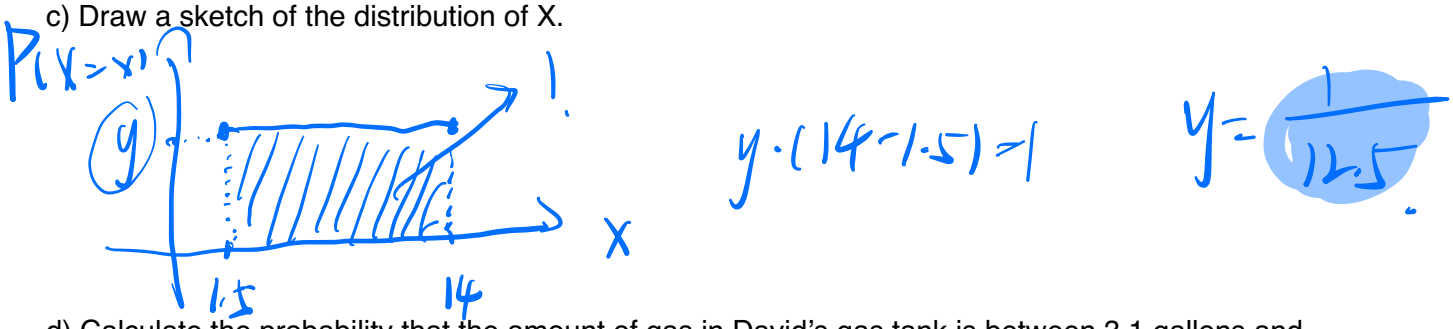
1. The amount of gas in David's gas tank is uniformly distributed between 1.5 and 14 gallons.

a) Define the random variable of interest, X.

b) State the distribution of X.

$X \sim \text{Uniform}(1.5, 14)$

c) Draw a sketch of the distribution of X.

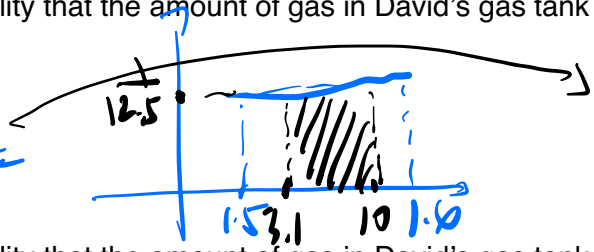


$y \cdot (14 - 1.5) = 1$

$y = \frac{1}{12.5}$

d) Calculate the probability that the amount of gas in David's gas tank is between 3.1 gallons and 10 gallons.

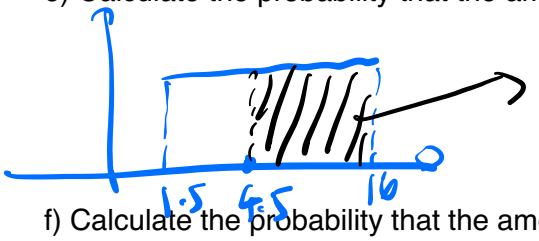
$P(3.1 \le X \le 10) =$



$\frac{1}{12.5} \times (10 - 3.1)$

$= 0.552$

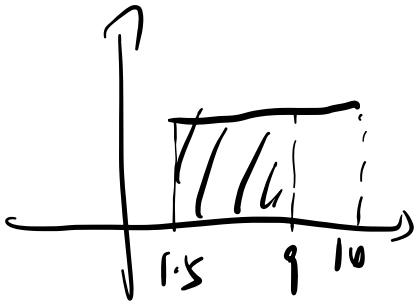
e) Calculate the probability that the amount of gas in David's gas tank is at least 4.5 gallons.



$\frac{1}{12.5} \times (14 - 4.5)$

$= 0.76$

f) Calculate the probability that the amount of gas in David's gas tank is less than 9 gallons.



$\frac{1}{12.5} \times (9 - 1.5) = 0.6$

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2. The wait time at a popular restaurant is uniformly distributed between twenty minutes and an hour.

a) Define the random variable of interest, X.

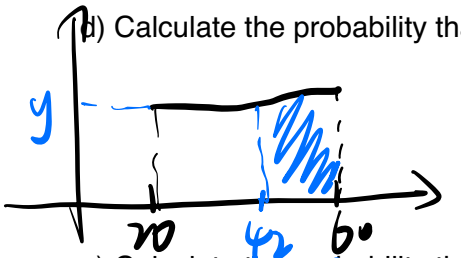
b) State the distribution of X.

Uniform(20, 60).

c) What is the average wait time at that restaurant?

$\frac{20+60}{2} = 40.$

d) Calculate the probability that the wait time is more than 42 minutes.



$y = \frac{1}{60-20} = \frac{1}{40}$

$P(X>42) = \text{Area} = \frac{1}{60} \times (60-42)$

$= 0.45.$

e) Calculate the probability that the wait time is at most 50 minutes.



$P(X \le 50) = \frac{1}{40} \times (50-20) = 0.75.$

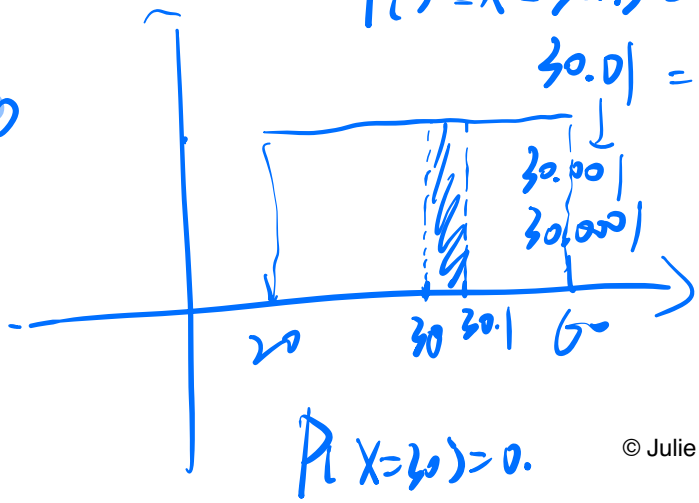
f) Calculate the probability that the wait time is 30 minutes.

$P(X=30) = 0$

$P(30 \le X \le 30.1) = \frac{1}{400}$

$30.01 = \frac{1}{1000}$

0.



$P(X=30) = 0.$

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3. The amount of honey harvested in a year from one of Jeff's bee colonies is normally distributed with a mean of 79 pounds and a standard deviation of 8.86 pounds.

a) Define the random variable of interest, X.

b) State the distribution of X.

$X \sim N(79, 8.86)$

c) What is the average amount of honey harvested in a year from one of Jeff's bee colonies?

$N(\mu, \sigma)$ μ . σ : . $\mu = 79$.

d) What is the probability that the amount of honey harvested in a year from one of Jeff's bee colonies is at least 70 pounds?

$\frac{70-\mu}{\sigma}$ $\frac{70-79}{8.86} = -1.02$

$P(X \geq 70) = P(Z \geq -1.02) = 1 - P(Z \leq -1.02) = 1 - 0.1539$

e) What is the probability that the amount of honey harvested in a year from one of Jeff's bee colonies is at most 75 pounds?

table $= 0.8461$

$\frac{75-79}{8.86} = -0.45$

0.1539

$P(X \leq 75) = P(Z \leq -0.45) = 0.3264$.

0.3264

f) What is the probability that the amount of honey harvested in a year from one of Jeff's bee colonies is between 79 and 90 pounds?

Calculate 2 Z values : a) $\frac{79-79}{8.86} = 0$

b) $\frac{90-79}{8.86} = 1.24$.

$P(79 \leq X \leq 90) = P(0 \leq Z \leq 1.24) = P(Z \leq 1.24) - P(Z \leq 0)$

$= 0.8925 - 0.5 = 0.3925$.

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